

Coproduct.

Def. Let  $\{X_i | i \in I\}$  be a collection of Topological Spaces.

Consider the set  $\coprod_{i \in I} X_i \stackrel{\text{def}}{=} \bigcup_{i \in I} X_i \times \{i\}$

and for each  $i \in I$ , Consider the injective map

$$\mathcal{Q}_i: X_i \rightarrow \coprod_{i \in I} X_i: x \mapsto (x, i).$$

The Topology  $\mathcal{Q}$  on  $\coprod_{i \in I} X_i$  which is the largest Topology such that

for all  $i \in I$ ,  $\mathcal{Q}_i$  is Continuous is called the disjoint union topology. Simply, denote  $\coprod X_i$ .

Prop. Let  $U \subset \coprod X_i$  be a subset of the disjoint union space. Then,

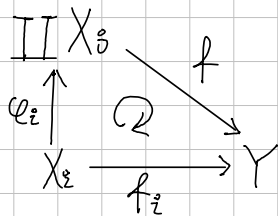
$U$  is open set of  $\coprod X_i$  if and only if for all  $i \in I$ ,  $\mathcal{Q}_i^{-1}[U] \subset X_i$  is open.

Proof. Since the Topology on  $\coprod X_i$  is the final Topology by  $\{\mathcal{Q}_i | i \in I\}$ , clear.  $\square$

Diagrammatically, for given a space  $Y$  and Continuous maps  $f_i: X_i \rightarrow Y$  ( $i \in I$ ), there exists a unique Continuous map

$$f: \coprod X_i \rightarrow Y: (x, i) \mapsto f_i(x)$$

which satisfies that for all  $i \in I$ ,  $f_i = f \circ \mathcal{Q}_i$ .



Adjunction space.

Def. Let  $X$  and  $Y$  be Topological spaces, and  $A \subset X$  be subspace. For a Continuous map  $f: A \rightarrow Y$ , define an equivalent relation on  $X \amalg Y$  which is generated by for all  $a \in A$ ,  $a \sim f(a)$ .

Then, the quotient topology

$X \cup_f Y \stackrel{\text{def}}{=} (X \amalg Y) / \sim$  is called the adjunction space by  $f$ .

And, the map  $f$  is called the attaching map.

Note: Specifically, the equivalent relation  $\sim$  on  $X \amalg Y$  is consisted by:

$$[z] = \begin{cases} f^{-1}[f[z]] \cup f[z] & \text{if } z \in A \\ \{z\} & \text{if } z \in X \setminus A \\ f^{-1}[f[z]] \cup \{z\} & \text{if } z \in f[A] \\ \{z\} & \text{if } z \in Y \setminus f[A] \end{cases}$$

In language of function, define the Canonical embeddings

$$\begin{aligned} \iota_X: X &\rightarrow X \cup_f Y & \text{and} & & \iota_Y: Y &\rightarrow X \cup_f Y \\ x &\mapsto (x, 0) & & & y &\mapsto (y, 1) \end{aligned}$$

$$\begin{array}{ccccc} X & \xrightarrow{\iota_X} & X \amalg Y & \xleftarrow{\iota_Y} & Y \\ & \searrow & \downarrow \pi & \swarrow & \\ & & X \cup_f Y & & \end{array}$$

Lemma. Let  $A \subset X$  be a closed (or open) set, and  $f: A \rightarrow Y$  be continuous. Then, the quotient map  $\pi: X \amalg Y \rightarrow X \cup_f Y$  induces a Homeomorphism

$$\pi|_Y: Y \xrightarrow{\cong} \pi[Y] \quad (\text{Canonical Copy})$$

Moreover,  $\pi[Y] \subseteq X \cup_f Y$  is a closed (or open) set.

$$\begin{array}{ccccc} X & \xhookrightarrow{\quad \iota_X \quad} & X \amalg Y & \xleftarrow{\quad \iota_Y \quad} & Y \\ & \searrow & \downarrow \pi & \swarrow & \\ & & X \cup_f Y & & \end{array}$$

Examples.

Mapping Cylinder.

Let  $f: X \rightarrow Y$  be a Continuous map. Then, for an interval  $I = [0, 1]$ ,  
let  $\tilde{f}: X \times \{1\} \rightarrow Y: (x, 1) \mapsto f(x)$ . Define the adjunction space

$$M_f \stackrel{\text{def}}{=} (X \times I) \cup_{\tilde{f}} Y$$